Interdependence and the Gains from Trade

Martin J. Conyon

A companion to the lecture slides. Each slide that hides a step is reproduced here and worked through line by line.

Before we begin

Required readings

Slide 2

Working through it

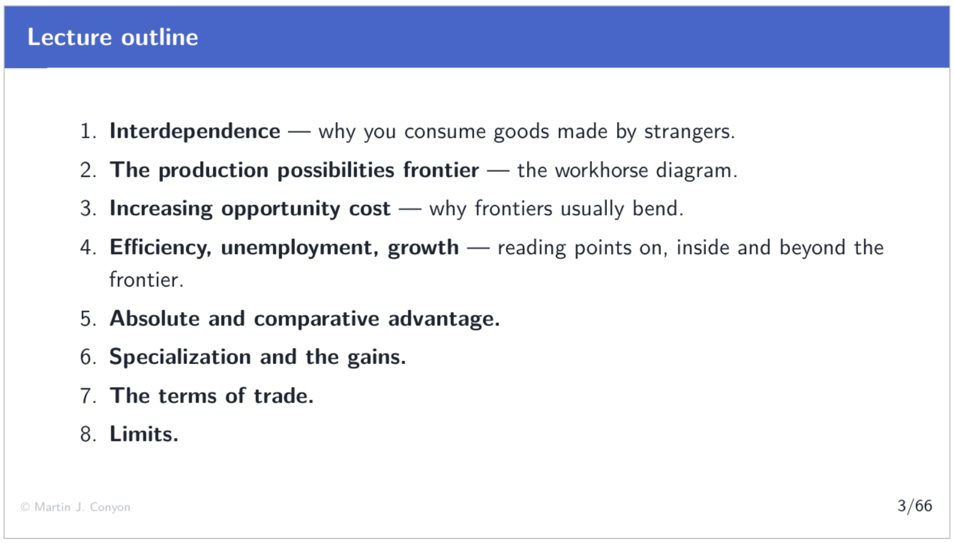
Why this chapter. Mankiw's chapter 3 runs the argument in the same order these slides do: build a frontier for each producer, read opportunity cost off the slopes, distinguish absolute from comparative advantage, and then find the range of prices at which both sides agree to trade.

Two differences worth knowing before you read. Mankiw writes productivity as *units per hour*; these slides follow Ricardo and write *hours per unit*. The two are reciprocals, and

Part 2 has a frame on exactly this. Mankiw also uses a farmer and a rancher rather than two countries — Part 8 explains why that substitution changes nothing.

How to use it. Read the chapter once before the material and once after. The slides move faster and leave out the worked detail; the text is slower and does not tell you which steps carry the weight.

Lecture outline



Slide 3

Working through it

How the eight blocks fit together. The order is not arbitrary, and half a minute spent on the shape makes the rest easier to hold.

Block 1 poses the puzzle: strangers who owe you nothing feed and clothe you, and one country can be better at everything and still trade.

Blocks 2–4 build the diagram that answers it. The frontier, what its slope means, why real frontiers bend, and how to read points on, inside and beyond it.

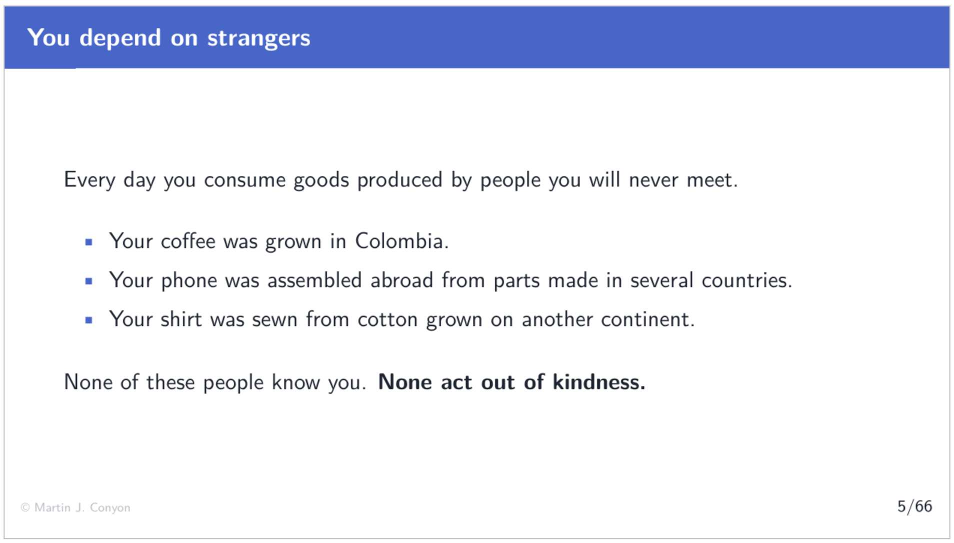
Blocks 5–7 are the argument itself: two senses of “better”, what specialization produces, and the price at which the two sides will actually agree.

Block 8 states what the model leaves out.

One thread runs through all of it. The slope of a frontier is an opportunity cost; two countries with different slopes have different opportunity costs; and wherever opportunity costs differ, both sides can gain. If you lose the thread, return to that sentence.

1. Part 1: Interdependence

You depend on strangers



Slide 5

Working through it

The point of opening here. Interdependence is so ordinary that it is invisible. The frame makes it visible by naming three objects you probably have within arm's reach.

Follow one of them. The coffee needed a grower, a picker, a processor, a shipper, a roaster, a retailer. Most of those people are in other countries. None of them has met you, and none of them will.

The bold line is the one that matters. Not one of them acts out of kindness. The grower is not thinking about your morning; he is thinking about his own income. That is not a cynical observation, it is the thing to be explained: how self-interested strangers end up supplying you reliably, cheaply, and without anyone coordinating it.

Hold the question open. The rest of the session is the answer.

Why does any of it reach you?

Slide 6

Working through it

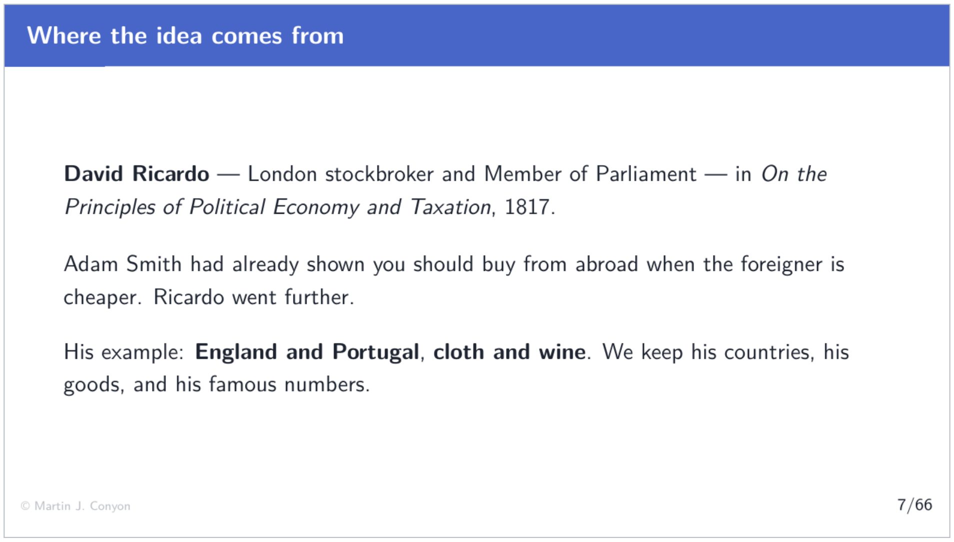
The short answer, and why it is not the whole answer. Both sides gain. A voluntary exchange happens only if each party expects to be better off, or one of them walks away. That much is easy.

Now the hard version. Two countries can both gain *even when one of them is better at producing everything.* Read the claim slowly, because it should not look obvious.

Why it should bother you. If Portugal makes cloth with fewer hours than England, and wine with fewer hours than England, then Portugal appears self-sufficient in a way England is not. England seems to have nothing to bring.

That intuition is wrong, and finding out exactly where it goes wrong is the work of this lecture. The answer is not effort, and it is not fairness. It is arithmetic.

Where the idea comes from



Slide 7

Working through it

Who Ricardo was. David Ricardo, a London stockbroker who made a fortune on the bond market and then entered Parliament, published *On the Principles of Political Economy and Taxation* in 1817.

What Smith had already established. If a foreign supplier can make something more cheaply than you can, buy it from them. Sensible, and not controversial.

What Ricardo added. That you should buy from abroad *even when you are the cheaper producer of everything* — provided the relative costs differ. That is the part that still surprises people two centuries later.

Why we keep his numbers. England and Portugal, cloth and wine, and the four hour-figures are Ricardo's own. Only the workforce sizes are supplied, and the next part says so explicitly. Working the original example means you are checking the argument that convinced economists, not a version tidied up afterwards.

2. Part 2: The Production Possibilities Frontier

Ricardo's data

Ricardo's data

England and Portugal each produce cloth and wine using **labor alone**.

Hours of labor needed per unit

	Cloth	Wine
England	100	120
Portugal	90	80

Read this carefully: these are costs. A bigger number means worse.

Slide 11

Working through it

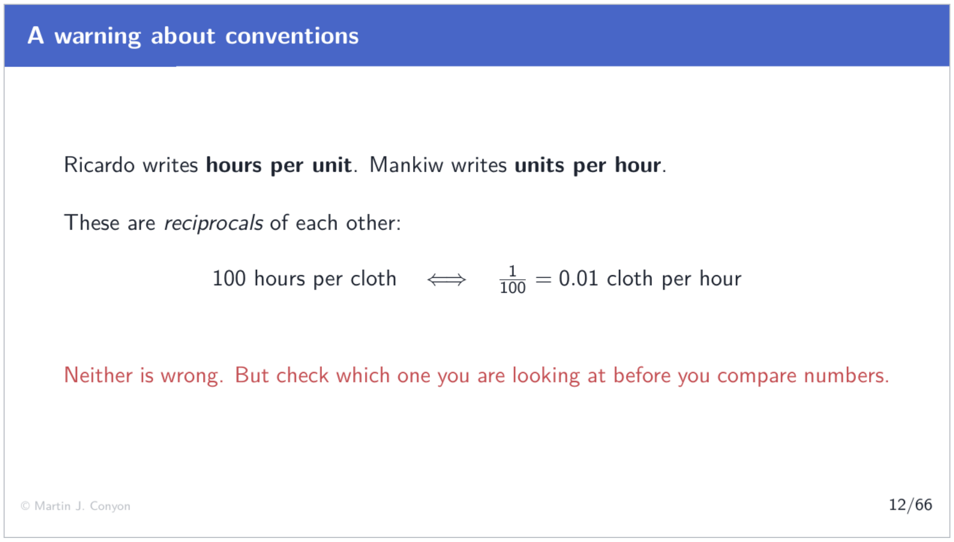
Read the table as costs. Every entry is the number of labor hours needed to make one unit. England: 100 hours per cloth, 120 per wine. Portugal: 90 per cloth, 80 per wine.

So a bigger number is worse. This is the reverse of the usual reflex, where a bigger number means more. Here, needing 120 hours for a wine is worse than needing 80.

Now compare the two rows. Portugal needs fewer hours than England for cloth (90 against 100) and fewer for wine (80 against 120). Portugal is more productive at both goods.

That is the difficulty, stated in four numbers. If Portugal is better at everything, the case for trade cannot rest on Portugal being unable to do something. It has to rest on something else, and Part 5 names it.

A warning about conventions



Slide 12

Working through it

Why this frame exists. It prevents an error that costs marks every year and that is invisible once it has been made, because the arithmetic still works — it just answers the wrong question.

Two conventions. Ricardo writes *hours per unit*: how much labor one unit costs. Mankiw writes *units per hour*: how much output one hour buys.

They are reciprocals. 100 hours per cloth is the same fact as $\frac{1}{100} = 0.01$ cloth per hour. Neither is more correct.

But they run in opposite directions. Under hours per unit, **smaller is better**. Under units per hour, **larger is better**. Mix them and every comparison in the session reverses: the country that looks more productive is actually less so.

The habit. Before comparing any two numbers, say out loud which convention the table is in. This deck is in hours per unit throughout.

How much labor does each country have?

Slide 13

Working through it

An honest gap in the source. Ricardo gave the productivity figures and nothing else. He never said how many workers England or Portugal had. Without that, there is no frontier to draw — productivity alone tells you the *slope* but not the *position*.

So the figures are supplied: England 3,600 hours, Portugal 2,880. Read that as ten workers against eight, each working the same year.

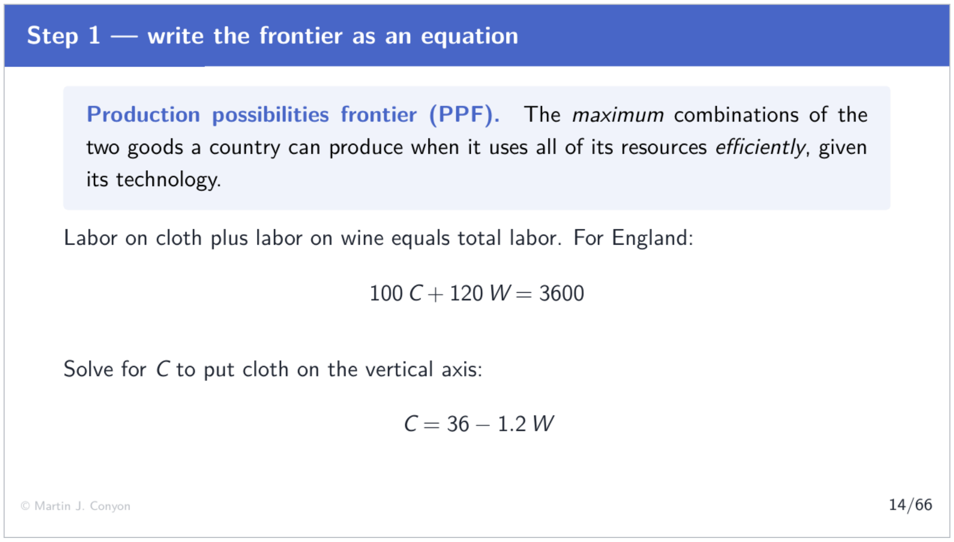
Why those two numbers. They are chosen so that every endpoint comes out whole:

$$3600/100 = 36, \qquad 3600/120 = 30, \qquad 2880/90 = 32, \qquad 2880/80 = 36.$$

Nothing in the economics depends on them.

What is invented and what is not. The workforce sizes are ours. The four productivity figures are Ricardo's, and those are the ones carrying the argument. Changing a workforce moves a frontier outward or inward; changing a productivity figure changes a slope, and the slopes are the whole result.

Step 1 — write the frontier as an equation



Slide 14

Working through it

Reading the definition. The frontier shows the *maximum* combinations of two goods available when resources are used fully and efficiently, given technology. Two conditions, and both do work: all the labor is used, and it is used well.

Where the equation comes from. Every hour goes to cloth or to wine, and on the frontier all of them are used. Hours on cloth are $100C$; hours on wine are $120W$; together they exhaust 3,600:

$$100C + 120W = 3600$$

Why rearrange. We want cloth on the vertical axis, so make C the subject. Subtract $120W$ from both sides, then divide throughout by 100:

$$100C = 3600 - 120W \implies C = 36 - 1.2\,W$$

Read the two numbers in the result. The 36 is cloth when $W = 0$ — all 3,600 hours on cloth at 100 hours each. The 1.2 is cloth lost per extra wine, and where that comes from gets a frame of its own shortly.

Nothing here is specific to frontiers. Write the constraint, rearrange, tabulate, plot. The same four steps handle any linear relationship.

Step 2 — plug the numbers in

Step 2 — plug the numbers in

England: $C = 36 - 1.2\,W$

Wine W	Substitute	Cloth C
0	$36 - 1.2(0)$	36
5	$36 - 1.2(5)$	30
10	$36 - 1.2(10)$	24
15	$36 - 1.2(15)$	18
20	$36 - 1.2(20)$	12
25	$36 - 1.2(25)$	6
30	$36 - 1.2(30)$	0

The ends: **36 cloth if England makes no wine; 30 wine if it makes no cloth.**

 15/66

Slide 15

Working through it

Build one row from scratch. Take $W = 15$:

$$C = 36 - 1.2(15) = 36 - 18 = 18$$

So 15 wine and 18 cloth. Check it directly against the constraint: $100(18) + 120(15) = 1800 + 1800 = 3600$. ✓

Every row is that same arithmetic, which is why the middle column shows the substitution rather than hiding it. You can put a finger on any entry and see where it came from.

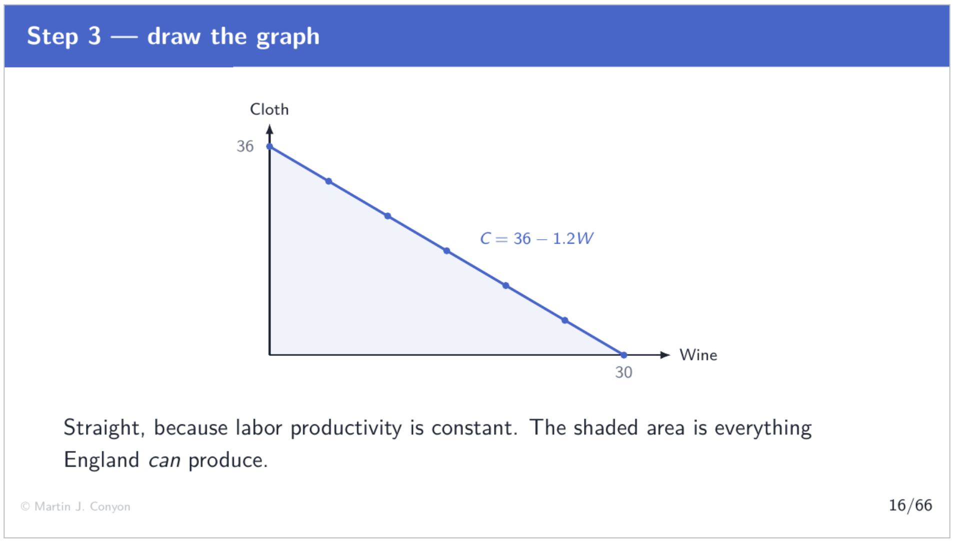
Now read down the cloth column: 36, 30, 24, 18, 12, 6, 0. Each step of 5 wine costs exactly 6 cloth — the same 6, every time.

That constancy is the thing to notice, and it is what makes the line straight. When the cost of the next unit never changes, the slope never changes.

The two endpoints. At $W = 0$, all hours go to cloth: 36 of them. At $W = 30$, all hours go to wine and cloth hits zero — check it, $120 \times 30 = 3600$, the whole supply.

And what lies between. Every mixture on the line, including fractional points the table does not list. Seven rows are a sample of a continuous relationship.

Step 3 — draw the graph



Slide 16

Working through it

What is on the axes. Wine horizontally, cloth vertically.

What the marks are. The dots are the seven rows of the table. The line is the function $C = 36 - 1.2W$. They coincide because they are the same object — the dots are the function at seven points, the line is the function everywhere.

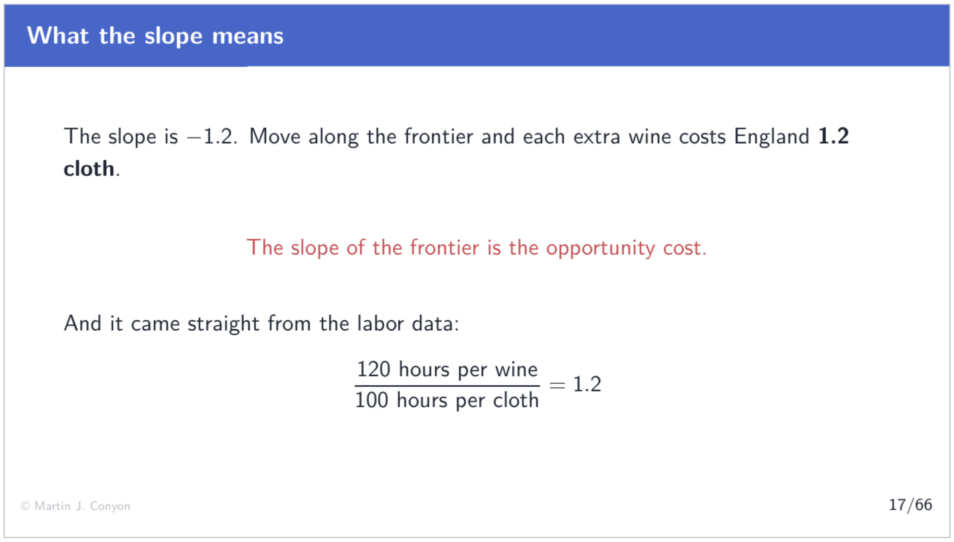
The two labelled ends are 36 on the vertical axis and 30 on the horizontal, exactly as computed.

What the shaded area means. Everything England *can* produce: the frontier itself, plus every point inside it. Points inside are feasible but wasteful — some hours idle or misused.

And what lies outside is the important part. Nothing there is available. Not expensive; unavailable, at any price, with this labor force and this technology.

Scarcity, drawn. The boundary exists because hours are finite. If labor were unlimited the shaded region would fill the page and there would be nothing to study.

What the slope means



Slide 17

Working through it

The central sentence of the block: the slope of the frontier is the opportunity cost. Not an analogy. The same number.

Reading it off the equation. In $C = 36 - 1.2W$, the coefficient on W is -1.2. Add one wine and cloth falls by 1.2.

And "cloth given up to get one more wine" is precisely the definition of opportunity cost — the value of the best alternative forgone. Here the alternative is cloth, and the forgone quantity is 1.2.

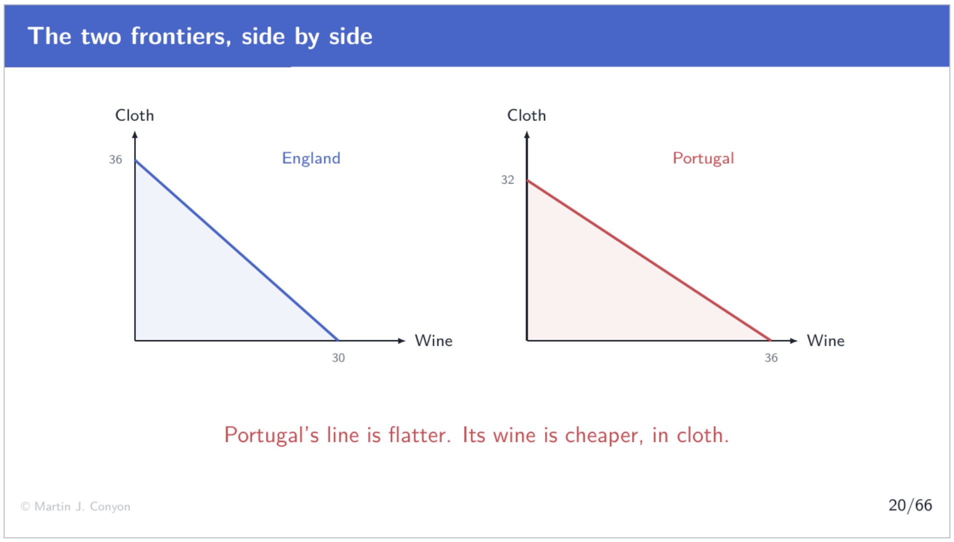
Where the 1.2 actually comes from. Not from the graph. From the labor data:

$$\frac{\text{120 hours per wine}}{\text{100 hours per cloth}} = 1.2$$

A wine absorbs 120 hours. Those 120 hours would otherwise have made 1.2 cloth, at 100 hours each. The geometry and the underlying technology are the same fact seen two ways.

Why the line is straight, and this is the assumption to notice. Every hour is as productive at one task as at the other, so it does not matter *which* hours move, and the cost of a wine is 1.2 cloth wherever you stand on the line. Part 3 removes that assumption, and the frontier stops being straight.

The two frontiers, side by side



Slide 20

Working through it

What to compare first. Neither frontier lies wholly inside the other. England reaches 36 cloth against Portugal's 32; Portugal reaches 36 wine against England's 30. Each country can out-produce the other in one good.

That is worth pausing on, because it is a consequence of the labor endowments we chose, not of the productivity figures. Portugal is more productive per hour at both goods and still cannot make more cloth, because it has fewer hours in total. Absolute advantage is about hours per unit; the reach of a frontier also depends on how many hours there are.

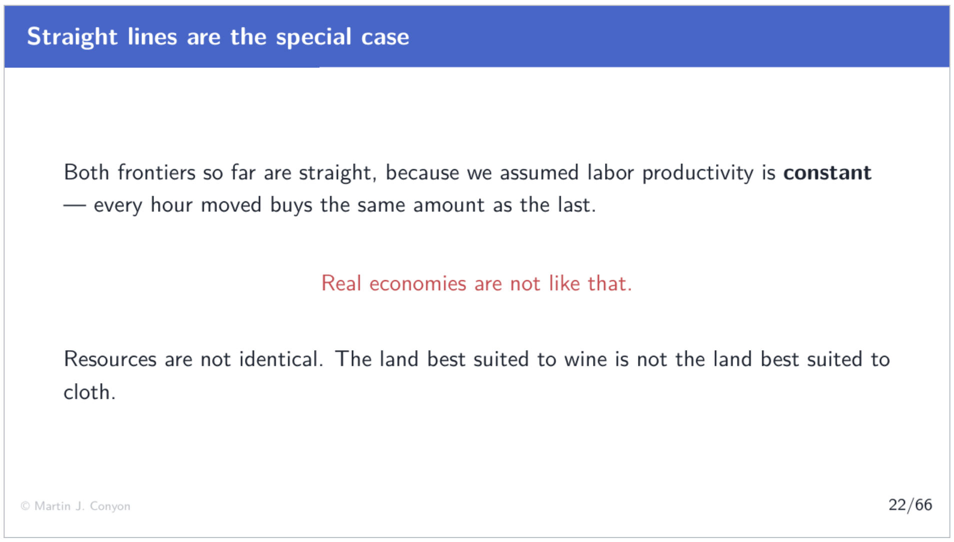
Now the steepness, which is what the session turns on. England's slope is -1.2; Portugal's is -0.89. Portugal's line is flatter.

What a flatter line means. Moving right along Portugal's frontier costs less cloth per wine gained. Wine is cheaper in Portugal — cheaper measured in cloth, which is the only measure available inside the model.

And a difference in slope is a difference in opportunity cost. Two countries, two different prices for the same trade-off. Part 5 turns that difference into the case for trade; everything before it is preparation.

3. Part 3: Increasing Opportunity Cost

Straight lines are the special case



Slide 22

Working through it

Naming the assumption that has been quietly doing the work. Both frontiers were straight because labor productivity was assumed *constant*: every wine always cost 120 hours and every cloth 100, regardless of which hours moved.

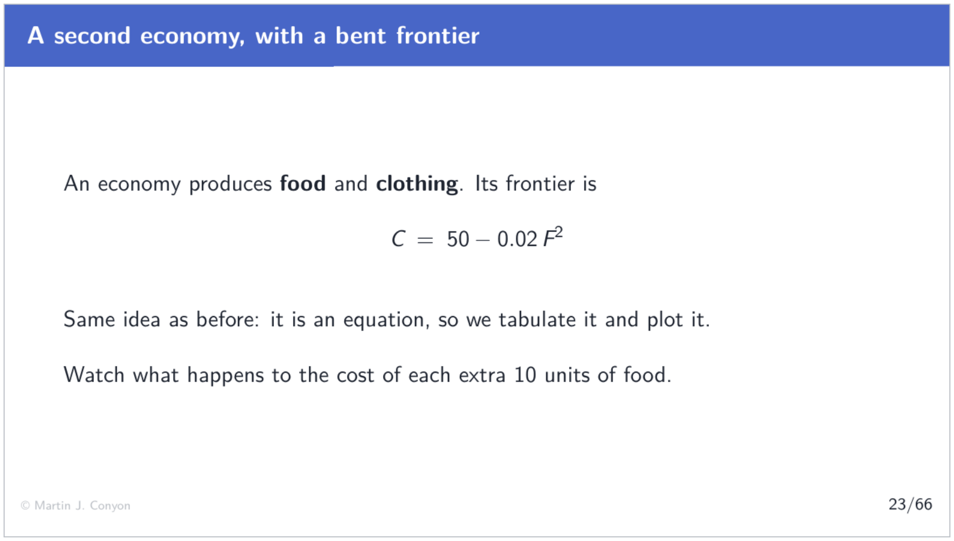
Why that made the lines straight. If all hours are identical, it does not matter which ones you move. The cost of the first wine and the thirtieth are the same, so the slope never changes, so the frontier is a line.

And real economies are not like that. Resources are specialized. The land best suited to vines is not the land best suited to sheep, and a weaver is not thereby a vintner.

So the order in which resources move starts to matter — and once order matters, the cost of the next unit depends on how much has already been moved, and the slope must change.

What this block does. A new economy, food and clothing, with specialized resources, worked through by exactly the method used before. The procedure does not change; only the shape of the answer does.

A second economy, with a bent frontier



Slide 23

Working through it

The reassurance this frame is offering. A curve looks harder than a line. The method does not change at all: it is an equation, so tabulate it and plot the table.

Reading the equation. $C = 50 - 0.02F^2$. Clothing is 50 minus something that grows with food.

Where the shape comes from. The F^2. Because food is squared, doubling food more than doubles the clothing sacrificed. That single term is the whole difference from the straight-line case, and you can see it before computing anything.

Check the two endpoints before going further. At $F = 0$: $C = 50$. At $C = 0$: $0.02F^2 = 50$, so $F^2 = 2500$ and $F = 50$. The frontier runs from 50 clothing to 50 food.

And the instruction in the last line is the point of the block. Do not watch the clothing column. Watch what each successive block of 10 food *costs*. The equation will not tell you the answer; the differences between rows will.

Plug the numbers in

Plug the numbers in

Food F	$C = 50 - 0.02F^2$	Clothing C	Cost of the last 10 food
0	$50 - 0.02(0)$	50	—
10	$50 - 0.02(100)$	48	2
20	$50 - 0.02(400)$	42	6
30	$50 - 0.02(900)$	32	10
40	$50 - 0.02(1600)$	18	14
50	$50 - 0.02(2500)$	0	18

2, 6, 10, 14, 18. Each extra 10 food costs more clothing than the last.

24/66

Slide 24

Working through it

Building one row. Take $F = 30$:

$$C = 50 - 0.02(30)^2 = 50 - 0.02(900) = 50 - 18 = 32$$

Note the order of operations — square first, then multiply by 0.02.

Where the last column comes from. It is the difference between consecutive clothing entries. From $F = 20$ to $F = 30$, clothing falls from 42 to 32, so that block of 10 food cost 10 clothing.

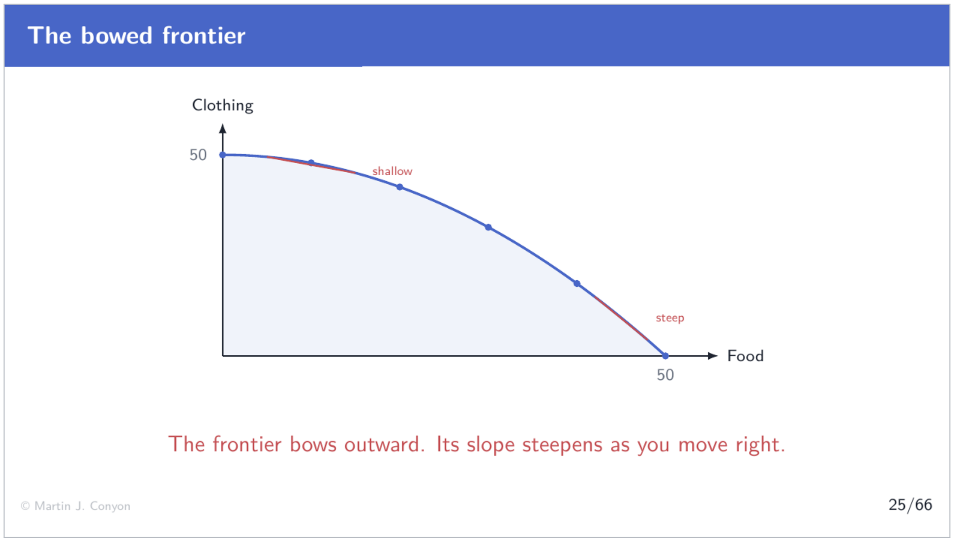
Now read that column: 2, 6, 10, 14, 18.

Each block of food costs more clothing than the one before it. That is **increasing opportunity cost**, and it is the entire content of this part.

Notice what did not happen. Nobody asserted increasing opportunity cost. It fell out of the equation, row by row, and every entry can be checked independently. The claim is demonstrated, not announced.

And the differences themselves form a pattern: 2, 6, 10, 14, 18 — rising by 4 each time. That regularity comes from squaring, and it is why a quadratic was chosen rather than some more complicated curve.

The bowed frontier



Slide 25

Working through it

What is on the axes. Food horizontally, clothing vertically.

What the marks are. The dots are the six rows of the table; the curve is $C = 50 - 0.02F^2$. They coincide, as before.

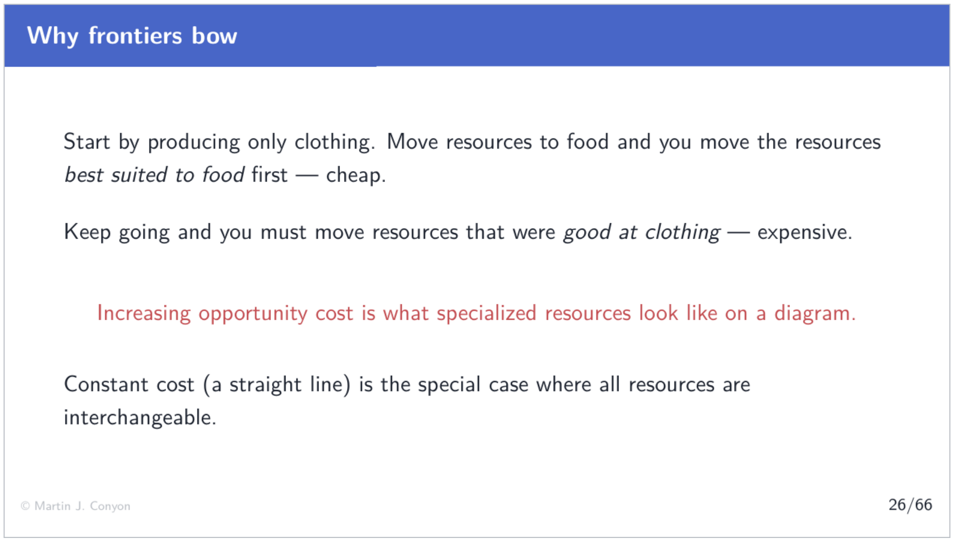
The shape: it bows outward*, away from the origin.* That word matters — outward, not upward or downward. The curve bulges away from the corner.

Reading the two red segments, which are the reason the figure exists. On the left, where little food is produced, the frontier is **shallow**: move right a long way and clothing barely falls. Food is cheap there.

On the right, the frontier is **steep**: the same rightward movement costs a great deal of clothing. Food is expensive there.

And the slope is still the opportunity cost — exactly the same interpretation as the straight-line case. The only difference is that the slope now changes as you move along the curve. One rule covers both shapes; in the straight case the slope simply happens to be constant.

Why frontiers bow



Slide 26

Working through it

The explanation is about the order *in which resources move.*

Start at the far left — all resources in clothing. Now move some into food. Which ones move first? The ones best suited to food and worst at clothing: the arable land, the people with agricultural skill. Moving them gains a lot of food and sacrifices very little clothing.

So the first block of food is cheap. Cost: 2 clothing, from the table.

Keep going and the pool of food-suited resources runs out. To expand food further you must move resources that were genuinely good at clothing — eventually the skilled weavers and the mills themselves.

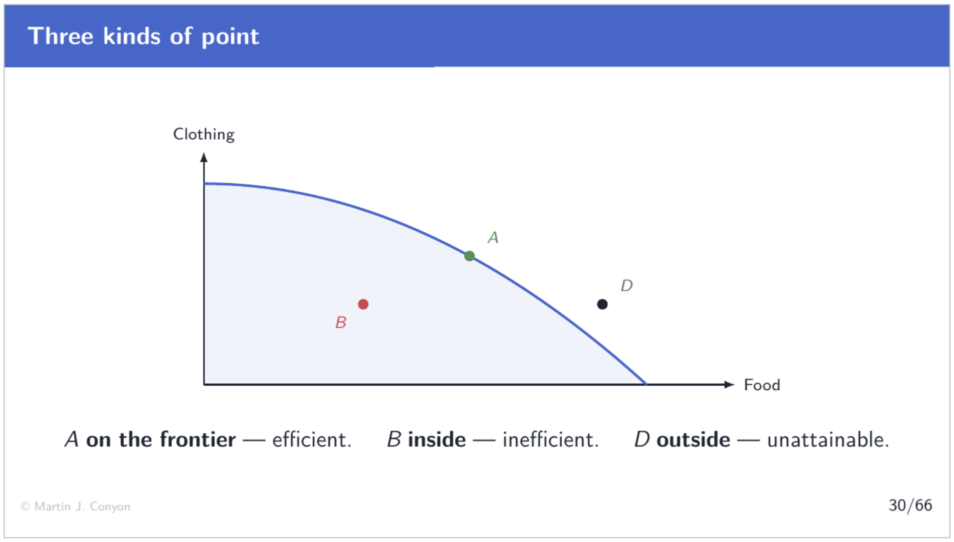
So the last block of food is dear. Cost: 18 clothing.

Which gives the sentence on the slide. Increasing opportunity cost is not a separate phenomenon requiring its own explanation. It is what specialized resources look like when you draw them.

And the argument runs both ways. Read the frontier from right to left and clothing behaves identically — cheap at first, dear later.

4. Part 4: Efficiency, Unemployment, Growth

Three kinds of point



Slide 30

Working through it

What is on the axes. Food and clothing, the same bowed frontier as the previous part.

The three points, and every question about a frontier is asking which kind you are looking at.

A, **on the frontier — efficient.** All resources are in use and used well. Getting more of one good requires giving up some of the other.

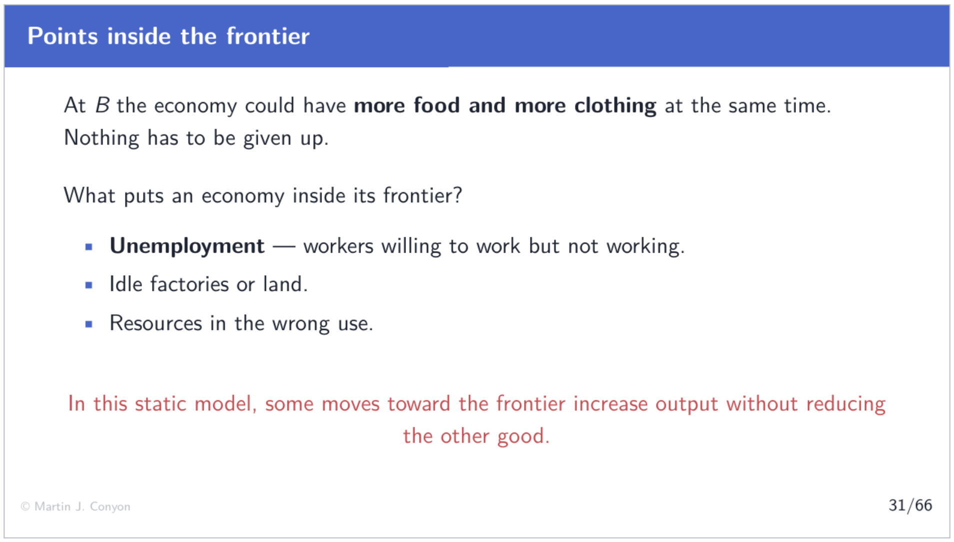
B, **inside — inefficient.** Feasible, but something is being wasted. More of *both* goods is available without giving anything up.

D, **outside — unattainable.** Not expensive. Not difficult. Simply not available with the resources and technology this economy has.

Note the asymmetry between B and D. B is a failure that can be fixed by using existing resources properly. D requires the frontier itself to move, which needs more resources or better technology — a different kind of change entirely.

The vocabulary to attach. Efficiency means being on the frontier. It says nothing whatever about *which* point on the frontier is best — that question involves preferences, and the diagram alone cannot answer it.

Points inside the frontier



Slide 31

Working through it

What is true at B that is not true anywhere on the frontier. The economy could have more food *and* more clothing at the same time. Nothing has to be given up.

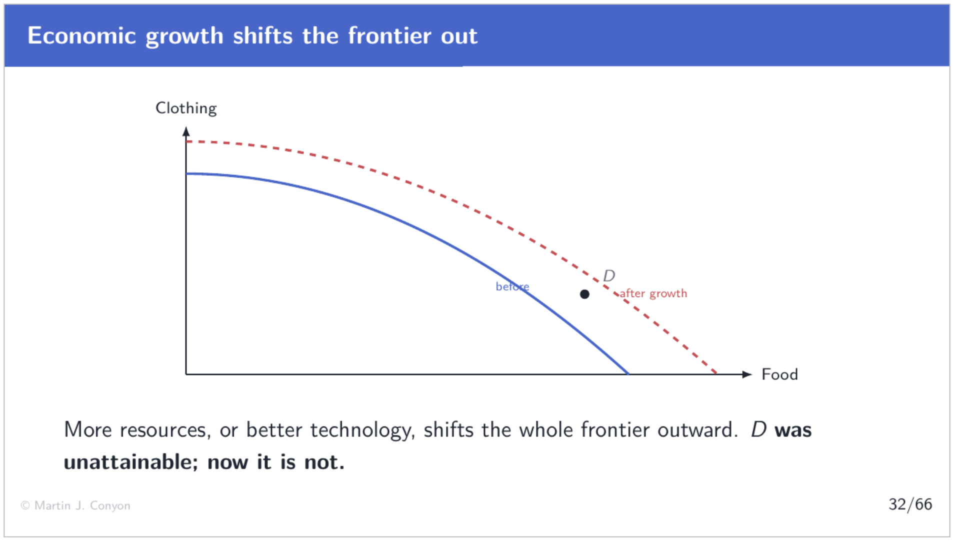
Which is a strange and important thing to say, because everything so far has been about trade-offs. Inside the frontier, the trade-off is suspended.

What puts an economy inside. **Unemployment** above all — workers willing to work and not working, so their hours produce nothing. Then idle factories or fallow land. Then resources in the *wrong* use, which is subtler: everything is employed, but not where it is most productive.

Note that the third case is invisible in unemployment statistics. An economy can be fully employed and still inside its frontier, if people are doing the wrong jobs.

Why the alert box is deliberately model-specific. From a strict interior point, at least one feasible move can raise one output without lowering the other. The diagram abstracts from search, matching, retraining and other adjustment costs, so it should not be read as saying that every route to the frontier is literally free.

Economic growth shifts the frontier out



Slide 32

Working through it

What the figure shows. Two frontiers. The solid one is the original; the dashed one lies outside it, after growth.

And point D is on the figure for a reason. It was the unattainable point two frames ago. Watch what happens to it.

Check D against both curves. D sits at 45 food and 20 clothing. On the *old* frontier, $C = 50 - 0.02(45)^2 = 50 - 40.5 = 9.5$. The economy could manage only 9.5 clothing at that quantity of food, and D needs 20. Out of reach. On the *new* frontier the same quantity of food permits about 25. D needs 20. Comfortably inside.

So the same point changed status without moving. D did not become cheaper or easier. The economy's capacity changed underneath it.

Which is what economic growth means on this diagram. Not sliding along a fixed boundary — that is a change of allocation, and it costs something. Moving the boundary itself, so combinations that were impossible become available.

And note the contrast with the previous frame. Reaching the frontier from inside is a matter of using what you already have. Moving the frontier outward is not: it requires more resources or better technology.

5. Part 5: Absolute and Comparative Advantage

Two measures of the cost of a good

Slide 36

Working through it

Two senses of "better", and the session has been building to the distinction.

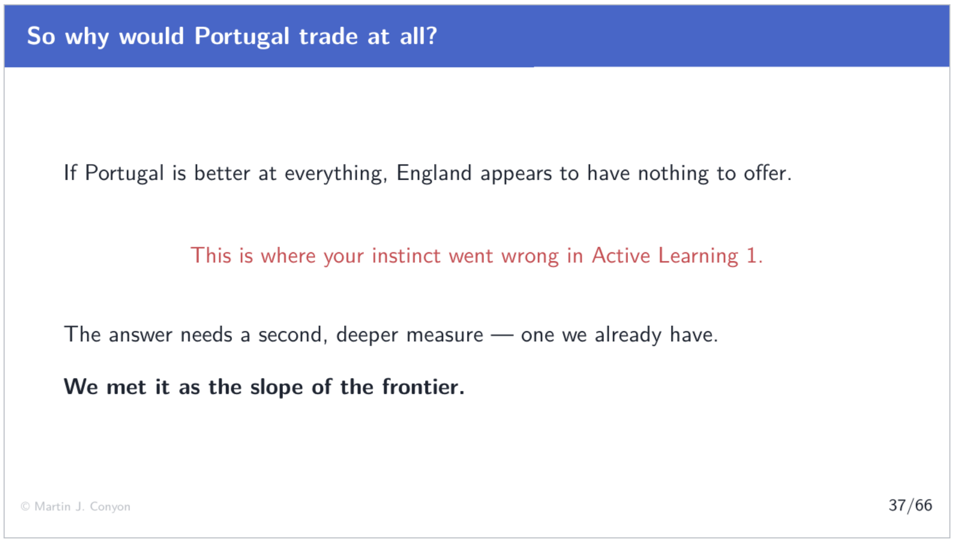
Reading the definition. Absolute advantage is producing a good using *fewer resources* — here, fewer labor hours — than the other country. It is a comparison *across* countries, one good at a time.

Apply it to Ricardo's table. Cloth: $90 < 100$, so Portugal. Wine: $80 < 120$, so Portugal. Portugal has the absolute advantage in both goods.

Why that seems to settle the matter. If Portugal can make anything with fewer hours, it appears to have no reason to buy from England, and England appears to have nothing to sell.

And that is precisely the wrong conclusion, which is what makes this the useful place to have introduced the term. Absolute advantage is a real measure of something — it just is not the measure that determines who trades with whom. The next two frames replace it.

So why would Portugal trade at all?



Slide 37

Working through it

The question this frame poses. Portugal is more productive at everything. England appears to have nothing to offer. So why would Portugal trade at all?

Where this was met before. Active Learning 1 asked exactly this, and most answers were A or B — Portugal should do everything, or Portugal gains and England loses. The instinct is that absolute productivity decides.

Why the instinct fails, stated without the machinery. Portugal has a fixed number of hours. Every hour it spends on cloth is an hour not spent on wine. Being good at cloth does not make those hours free; it only makes them productive. Something is still given up.

So the relevant question is not "who makes it with fewer hours?" but "who gives up less to make it?" Those are different questions with different answers.

And the second one has already been answered. It is the slope of the frontier: opportunity cost. The tool needed to solve the puzzle was built in Part 2, before the puzzle was posed.

Opportunity cost, for both countries

Opportunity cost, for both countries

Hours spent on one good are hours taken from the other:

$$\text{opportunity cost of a good} = \frac{\text{hours that good needs}}{\text{hours the other good needs}}$$

	1 wine costs (cloth)	1 cloth costs (wine)
England	$120 \div 100 = 1.20$	$100 \div 120 = 0.83$
Portugal	$80 \div 90 = 0.89$	$90 \div 80 = 1.125$

Within each country the two columns are reciprocals. **If one good is dear, the other must be cheap.**

 38/66

Slide 38

Working through it

Where the formula comes from. Labor is the only input, so hours spent on one good are hours taken from the other. Making one wine costs 120 hours, and those hours would otherwise have made cloth at 100 hours each. Hence

$$\text{opportunity cost of a good} = \frac{\text{hours that good needs}}{\text{hours the other good needs}}$$

Working the four entries.

- England, 1 wine: $120 \div 100 = 1.20$ cloth.
- England, 1 cloth: $100 \div 120 = 0.83$ wine.
- Portugal, 1 wine: $80 \div 90 = 0.89$ cloth.
- Portugal, 1 cloth: $90 \div 80 = 1.125$ wine.

Within each row the two entries are reciprocals. $1.20 \times 0.83 \approx 1$ and $0.89 \times 1.125 \approx 1$. That is forced, not coincidental: with two goods and one resource, the cost of X in terms of Y and the cost of Y in terms of X are two readings of a single ratio.

Which gives a claim worth stating plainly. If one good is dear, the other *must* be cheap. A country cannot find both goods expensive in terms of each other — the statement is not merely false, it is incoherent.

And notice where you have seen 1.20 and 0.89 before. They are the two frontier slopes from Part 2. The table is the same information in a different layout.

Comparative advantage

Slide 39

Working through it

Reading the definition. Comparative advantage is producing a good at a *lower opportunity cost* than the other country. Compare the same good across countries, using the opportunity-cost figures rather than the hours.

Wine. Portugal gives up 0.89 cloth; England gives up 1.20. Portugal sacrifices less, so **Portugal has the comparative advantage in wine**.

Cloth. England gives up 0.83 wine; Portugal gives up 1.125. England sacrifices less, so **England has the comparative advantage in cloth**.

Now look at what just happened. Portugal makes cloth with fewer hours than England — 90 against 100 — and England still has the comparative advantage in cloth. Absolute advantage and comparative advantage have come apart, and it is the second one that governs trade.

Why they come apart. Portugal is better at both, but it is *most* better at wine: 80/120 of England's hours for wine, against 90/100 for cloth. Putting a Portuguese hour into cloth wastes more of Portugal's particular talent than putting it into wine does.

The sentence to carry. Being better at something in absolute terms does not mean you should do it. What matters is what you give up.

The key insight

Slide 40

Working through it

The claim. Unless two countries have *identical* opportunity costs, each has a comparative advantage in something.

Why it is forced rather than usual. Within a country the two opportunity costs are reciprocals — they multiply to one. Suppose a country had the lower opportunity cost in *both* goods. Lower in wine means its wine-in-cloth figure is smaller than the other country's. But its cloth-in-wine figure is the reciprocal of that, so it must be *larger* than the other country's. The supposition contradicts itself.

So comparative advantage in both goods is arithmetically impossible. Not unlikely. Impossible.

Which gives the result on the slide. However unproductive a country is, there is always some good it gives up least to produce, and therefore always a basis for trade. No country is ever priced out of the world economy altogether.

The one exception, and it is the boundary case. If the two countries have exactly the same opportunity costs, neither has an advantage in either good and there is nothing to gain. Part 7 shows what that looks like: the band of workable prices closes to a point.

6. Part 6: Specialization and the Gains

Life without trade

Slide 44

Working through it

Reading the definition. Autarky is the no-trade case: a country consumes exactly what it produces. The word is worth knowing because it names the baseline every gain is measured against.

What autarky does to the diagram. Without trade, a country's production frontier is also its *consumption* frontier. Whatever it eats or drinks, it must first have made. Points beyond the frontier are unavailable, full stop.

Why a starting point has to be assumed. The frontier is a whole line, and the country could sit anywhere on it. To measure a gain we need one specific point, so we choose one: an equal split of labor between the two goods. England 1,800 hours on each; Portugal 1,440 on each.

Note that this is a choice, not a result. No argument says a country would split its labor evenly. It is picked because it is neutral — it favours neither good and neither country.

And nothing turns on it. Any point on each frontier would do. The gains that follow come from the difference in slopes, not from where the two countries happened to start.

Autarky: production equals consumption

Autarky: production equals consumption

	Cloth	Wine
England	$1800/100 = 18$	$1800/120 = 15$
Portugal	$1440/90 = 16$	$1440/80 = 18$
World	**34**	**33**

Without trade the world produces **34 cloth and 33 wine**. Remember those two numbers.

 45/66

Slide 45

Working through it

Every entry is a division, and the table shows it. With 1,800 hours on cloth at 100 hours each, England makes $1800/100 = 18$ cloth. With 1,800 hours on wine at 120 hours each, $1800/120 = 15$ wine.

Portugal, the same way. $1440/90 = 16$ cloth and $1440/80 = 18$ wine.

Check each pair against its frontier. England at $(W, C) = (15, 18)$: $36 - 1.2(15) = 18$. ✓ Portugal at $(18, 16)$: $32 - 0.89(18) \approx 16$. ✓ Both countries are on their frontiers, as autarky requires.

Now the world totals. Cloth: $18 + 16 = 34$. Wine: $15 + 18 = 33$.

Why those two numbers matter more than any others in the session. Every claim about gains from here on is a comparison against 34 and 33. If the world after specialization has more than that, the gain is real and it is measurable. If not, there is nothing to argue about.

Now specialize

Now specialize

Each country moves its labor into its comparative-advantage good: England into cloth, Portugal into wine.

		Cloth	Wine
England	(3600/100)	36	0
Portugal	(2880/80)	0	36
World		**36**	**36**

Cloth 34 → 36. Wine 33 → 36.

 46/66

Slide 46

Working through it

What "specialize" means here. Each country moves *all* its labor into the good in which it has the comparative advantage. England into cloth, Portugal into wine — the assignment Part 5 established.

The arithmetic. England: $3600/100 = 36$ cloth, and no wine. Portugal: $2880/80 = 36$ wine, and no cloth.

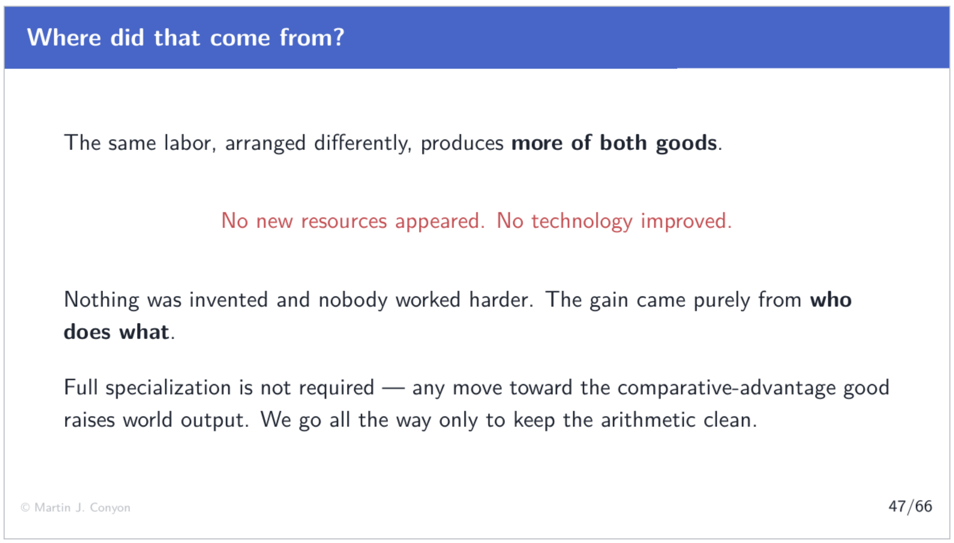
World totals now: 36 cloth and 36 wine.

Set that beside autarky. Cloth $34 \to 36$. Wine $33 \to 36$. Two more cloth and three more wine, from the same workers, the same hours, and the same technology.

Note which country makes which good. Portugal makes the wine even though it is better at cloth than England is. That is comparative advantage doing the allocating rather than absolute advantage, and the world total is the evidence that it allocates better.

One caution about the word "all". Full specialization is a convenience, not a requirement. Any move toward the comparative-advantage good raises world output; going the whole way simply keeps the arithmetic clean.

Where did that come from?



Slide 47

Working through it

The question the frame poses. Two more cloth and three more wine appeared. Where from?

Rule out the obvious answers, one at a time. No new workers were hired — the hours are 3,600 and 2,880, exactly as before. No new machines, no new land. No technique improved: England still needs 100 hours for a cloth, Portugal still needs 80 for a wine. Nobody worked longer, and nobody worked harder.

So the inputs are identical and the output is larger. The only thing that changed is *who does what.*

Why that is more than a curiosity. Under autarky, England spent 1,800 hours on wine at 120 hours each — its worst use of labor, by its own opportunity costs. Portugal spent 1,440 hours on cloth, likewise. Specialization stops both countries doing the thing they are relatively bad at. The gain is the waste that was removed.

And the practical point in the last line. Full specialization is not required. Any reallocation *toward* the comparative-advantage good raises world output, which matters once frontiers bow and complete specialization stops being optimal.

But now there is a problem

Slide 48

Working through it

The problem specialization creates. England now holds 36 cloth and no wine. Portugal holds 36 wine and no cloth. Each country has more of one thing than it could ever want and none of the other.

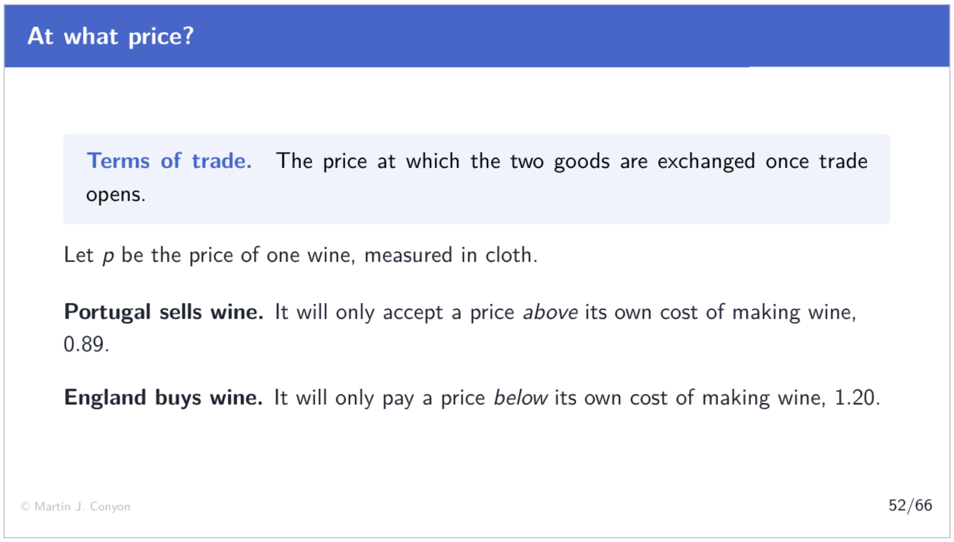
So a larger world output is not yet a gain to anybody. Production rose; consumption has not. England cannot drink cloth.

Which is why exchange is not an optional extra to this argument. The gain exists only if the extra output can be redistributed so that both countries end up with a balanced basket. Specialization without trade would leave both worse off than autarky.

And that raises the question the next block answers. They will swap — but at what rate? One cloth for one wine? Two for one? The answer is not arbitrary, and it is not a matter of bargaining skill alone. It is bounded on both sides by the two opportunity costs already computed.

7. Part 7: The Terms of Trade

At what price?



Slide 52

Working through it

Reading the definition. The terms of trade is the price at which the two goods exchange once trade opens. Here we measure it as p, the price of one wine in units of cloth.

Note that there is no money in this model. A price is a ratio between two goods, nothing more. "$p = 1$" means one cloth buys one wine.

The seller's condition. Portugal sells wine. It can make its own cloth by giving up 1.125 wine per cloth — equivalently, its wine is worth 0.89 cloth to it. If the world offers less than 0.89 cloth per wine, Portugal does better keeping the wine and making cloth itself. So Portugal requires $p > 0.89$.

The buyer's condition. England buys wine. It can make its own wine at a cost of 1.20 cloth each. If the world charges more than 1.20, England does better making its own. So England requires $p < 1.20$.

Notice what both conditions have in common. Each country compares the world price against its *own* opportunity cost — the number computed in Part 5. Neither is asking what is fair. Each is asking whether the deal beats its alternative.

The band

The band

$$0.89 \ < \ p \ < \ 1.20 \qquad \text{(cloth per wine)}$$

A deal exists only in the gap between the two opportunity costs.

Any price in that band benefits both. We take a round number in the middle, $p = 1$ — one cloth for one wine.

The price decides how the gains are split. It does not decide whether they exist.

 53/66

Slide 53

Working through it

Putting the two conditions together. Portugal needs $p > 0.89$; England needs $p < 1.20$. Both hold simultaneously only in the band

$$0.89 < p < 1.20$$

What the band is made of. Its two edges are the two countries' opportunity costs. Nothing else determines them.

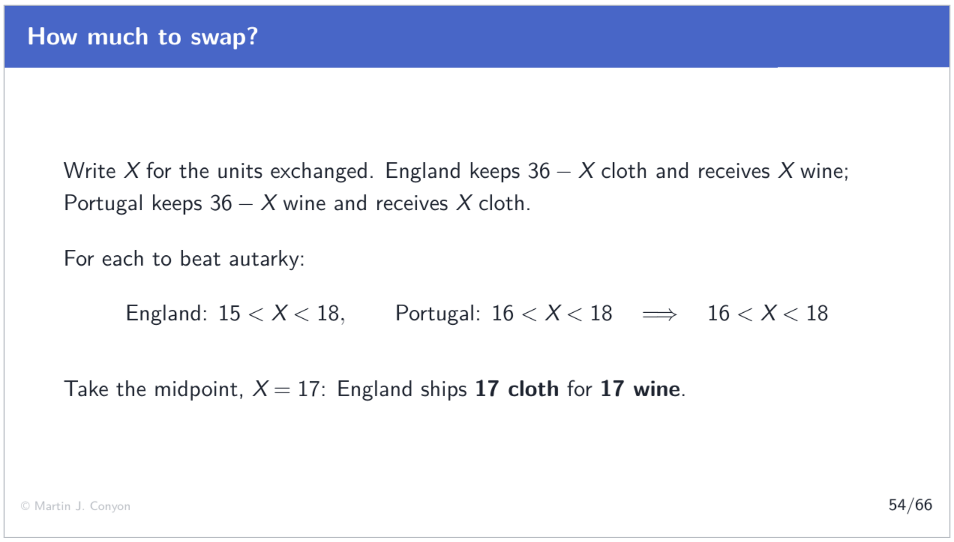
So the width of the band measures how different the two countries are. Here it is about 0.31 cloth per wine. The more the opportunity costs differ, the wider the band and the more room there is to strike a deal.

And the limiting case is the informative one. If both countries had the same opportunity cost, the band would close to a single point and there would be no price strictly better than autarky for both. **Differing opportunity costs are not merely helpful to trade; they are the reason it happens.**

Why $p = 1$. It sits inside the band and keeps the arithmetic clean. Any other price in the band would also work.

The distinction in the last line, and it is worth holding. Where the price sits inside the band determines *how the gains are divided* between the two countries. Whether a gain exists at all is settled by the band being non-empty.

How much to swap?



Slide 54

Working through it

Setting up the quantity. At $p = 1$ the swap is one for one, so a single number X describes it: England ships X cloth and receives X wine, and Portugal the reverse.

England's condition. After trade England holds $36 - X$ cloth and X wine. Under autarky it had 18 cloth and 15 wine. To beat autarky on both goods it needs $36 - X > 18$ and $X > 15$, that is $15 < X < 18$.

Portugal's condition. After trade Portugal holds X cloth and $36 - X$ wine, against 16 and 18 under autarky. So $X > 16$ and $36 - X > 18$, that is $16 < X < 18$.

Both together. The binding constraints are Portugal's lower bound and the shared upper bound: $16 < X < 18$.

Note the shape of the argument. Exactly as with the price, each side is compared against its own autarky position, and the workable region is the overlap. The quantity is bounded for the same reason the price is.

Take the midpoint, $X = 17$*,* which is inside the range and gives whole numbers throughout.

After trade

After trade

	Cloth made	Cloth eaten	Wine made	Wine drunk
England	36	$36 - 17 = 19$	0	$0 + 17 = 17$
Portugal	0	$0 + 17 = 17$	36	$36 - 17 = 19$
World	**36**	**36**	**36**	**36**

Each country now consumes a balanced basket it could not have produced alone.

55/66

Slide 55

Working through it

Reading the table one row at a time. England produces 36 cloth, ships 17, and consumes $36 - 17 = 19$. It produces no wine, receives 17, and consumes $0 + 17 = 17$.

Portugal is the mirror image. It produces 36 wine, ships 17, consumes 19; produces no cloth, receives 17, consumes 17.

Check the world columns. Cloth made 36, cloth consumed $19 + 17 = 36$. Wine made 36, wine consumed $17 + 19 = 36$. Nothing appears and nothing vanishes — trade redistributes what specialization produced.

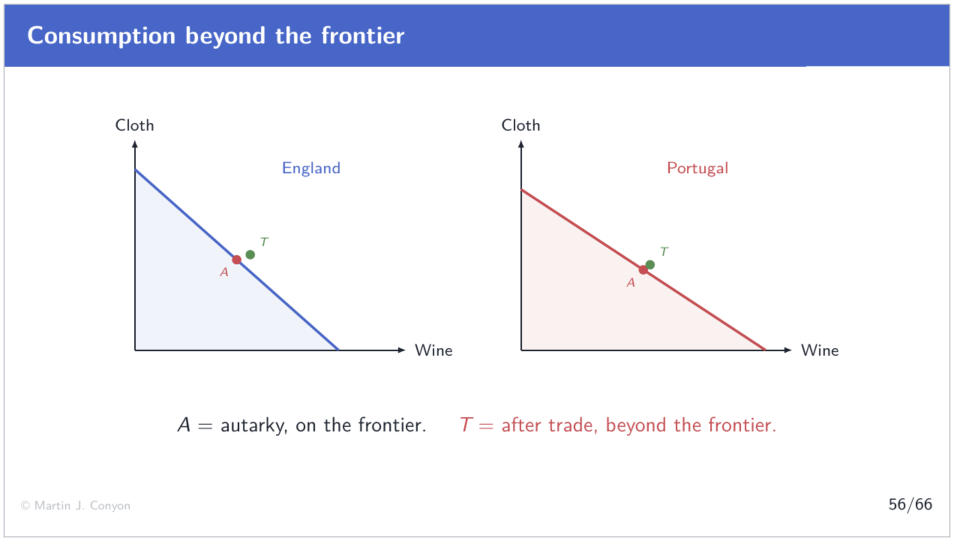
Now set consumption against autarky.

- England: 18 cloth and 15 wine → **19 cloth and 17 wine**.
- Portugal: 16 cloth and 18 wine → **17 cloth and 19 wine**.

Both countries have more of both goods. Not more of one at the expense of the other, and not more for one country at the expense of the other. Strictly more of everything, for both.

And each basket is balanced, which is what neither country could have achieved alone at these quantities. That is the claim the next figure draws.

Consumption beyond the frontier



Slide 56

Working through it

What the two figures show. Each country's own production frontier, with two points marked on it.

A **is autarky** — 15 wine and 18 cloth for England, 18 wine and 16 cloth for Portugal. Both sit *on* the frontier, as autarky requires: a country consumes exactly what it produces, and production must be feasible.

T **is after trade** — 17 and 19 for England, 19 and 17 for Portugal. Both sit *outside* the frontier.

That should stop you for a moment. Part 2 established that points beyond the frontier are unattainable, and Part 4 repeated it. Now a point beyond the frontier is being consumed.

The resolution is a distinction the diagram has been carrying all along. The frontier bounds what a country can *produce*. It does not bound what a country can *consume*. Those coincide only under autarky, which is exactly why autarky was defined before this figure was drawn.

So trade does not repeal the constraint. England still produced on its frontier — at the corner, 36 cloth. What trade did was convert cloth into wine at a better rate than England's own technology could manage.

Which is the geometric statement of the whole session. Production stays inside the boundary. Consumption escapes it.

The gains, side by side

The gains, side by side

	Autarky		Trade	
	Cloth	Wine	Cloth	Wine
England	18	15	**19**	**17**
Portugal	16	18	**17**	**19**

Both countries consume more cloth *and* more wine.

Answer C was right. Trade is not a contest with a winner and a loser.

© Martin J. Conyon 57/66

Slide 57

Working through it

The table is the argument's balance sheet, and it is worth reading as one.

England: 18 and 15 becomes 19 and 17. One more cloth, two more wine.

Portugal: 16 and 18 becomes 17 and 19. One more cloth, one more wine.

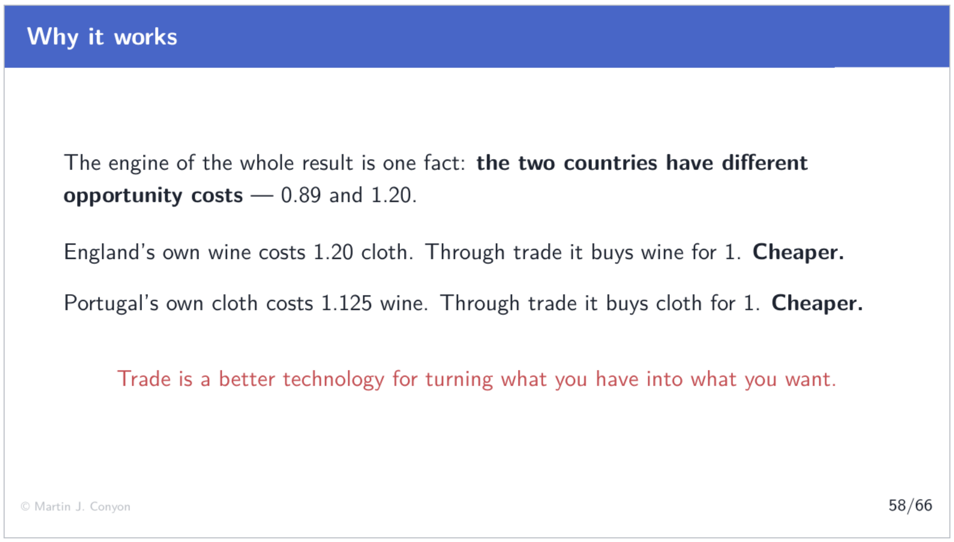
Note what is absent. No negative number anywhere. Nobody trades away a good and ends with less of it. This is not a redistribution in which one country's gain is the other's loss.

And note what produced it. The same 3,600 and 2,880 hours, the same productivity figures, the same technology. Only the allocation of labor and the subsequent exchange changed.

Which settles Active Learning 1. Answer C — both gain — and the table is the demonstration rather than an assertion. Most classes predict A or B because the instinct is that the more productive country must win. It does not win, because there was never a contest.

One honest limit on the claim. The table shows both *countries* gaining. It says nothing about who inside each country gains, and Part 8 takes that up.

Why it works



Slide 58

Working through it

The engine, stated once. The two countries have different opportunity costs — 0.89 and 1.20. Everything else in the session is machinery for exploiting that difference.

Watch the gain arise on England's side. Making its own wine costs England 1.20 cloth per wine. Buying wine through trade costs 1 cloth per wine. England acquires wine at a lower price than its own technology can manage.

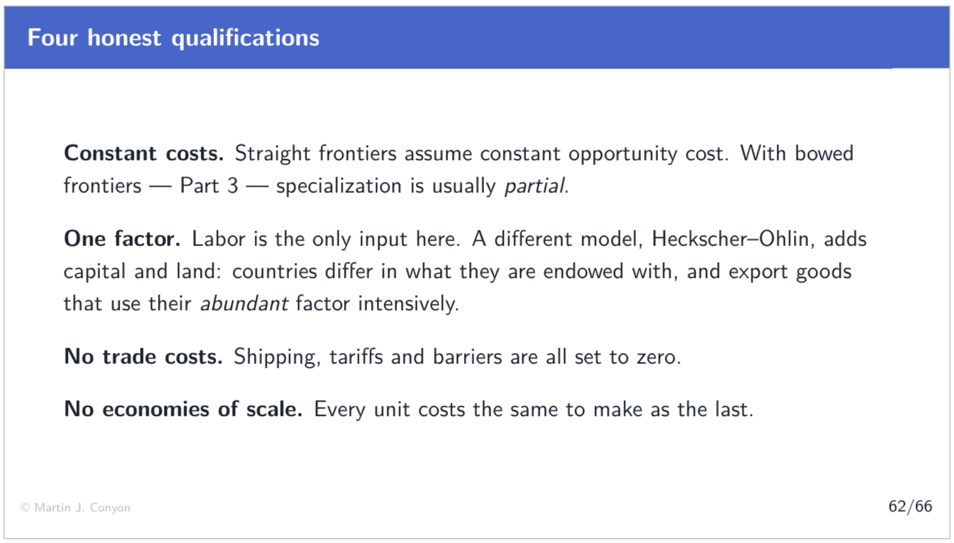
And on Portugal's side. Making its own cloth costs Portugal 1.125 wine per cloth. Buying cloth through trade costs 1 wine per cloth. Also cheaper.

Both sides buy below their own cost of production, simultaneously. That sounds impossible if you think of cost as an absolute quantity. It is straightforward once cost means *what you give up*, because the two countries give up different things.

Reading the sentence in the alert box. Trade is a technology for turning what you have into what you want. England has a way of turning cloth into wine without trade — move labor along its frontier, at 1.20 cloth per wine. Trade is a second way of doing the same thing, at 1. It is simply the better machine.

8. Part 8: Limits and Extensions

Four honest qualifications



Slide 62

Working through it

Why a block of qualifications belongs here rather than nowhere. The result is strong and it is easy to overstate. Each of the four assumptions below is doing work, and each fails somewhere in the real world.

Constant costs. Straight frontiers assume constant opportunity cost. With the bowed frontiers of Part 3, the cost of the export good rises as a country specializes, and specialization typically stops *partway* — both countries still produce both goods. The direction of the result survives; the corner solution does not.

One factor. Labor is the only input here, so productivity differences have no explanation — they are simply given. Heckscher–Ohlin is a different model, not an extension: countries are endowed with different amounts of capital, land and labor, and export the goods that use their abundant factor intensively.

No trade costs. Shipping, tariffs, insurance and border delays are all set to zero.

No economies of scale. Every unit costs the same to make as the last. Where scale economies are large, a country can acquire an advantage by producing a lot of something, which reverses the direction of causation in this model.

The reason for listing them. A model whose limits you cannot state is a model you do not understand.

Winners and losers inside a country

Slide 63

Working through it

The gap between two true statements. The nation as a whole gains. Specific people inside it lose. Both follow from the same analysis, and the model as built only shows the first.

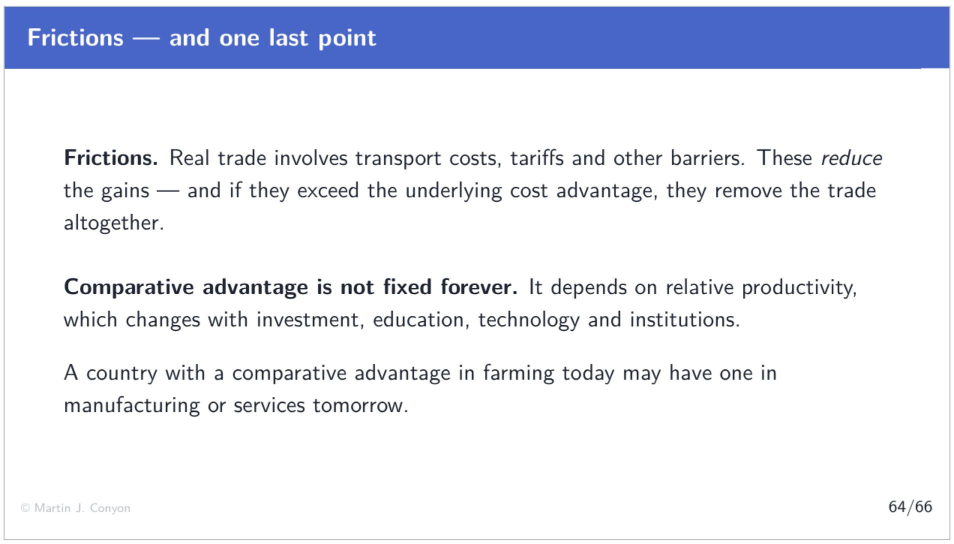
Trace the mechanism. Opening to trade moves labor into the comparative-advantage industry and out of the import-competing one. In the model that reallocation is instantaneous and costless: a worker simply appears in the other sector.

In reality it is neither. Workers in the contracting industry face unemployment, relocation, and retraining. Skills built over a career in one sector may transfer poorly. The costs fall on identifiable people, in identifiable places, at a definite time.

Note that this is not an objection to the arithmetic. The gains are real and they are larger than the losses — that is what "the nation gains" means. The difficulty is distributional: the gains are diffuse and the losses are concentrated.

Which explains a political puzzle worth naming. Economists agree on the gains from trade to an unusual degree, and trade policy is fought over continuously. Both facts are consistent, because the question "does the country gain?" and the question "should we do it?" are not the same question.

Frictions — and one last point



Slide 64

Working through it

Frictions. Transport, tariffs, insurance and regulatory barriers all reduce the gains, and the arithmetic of when they eliminate them is worth doing.

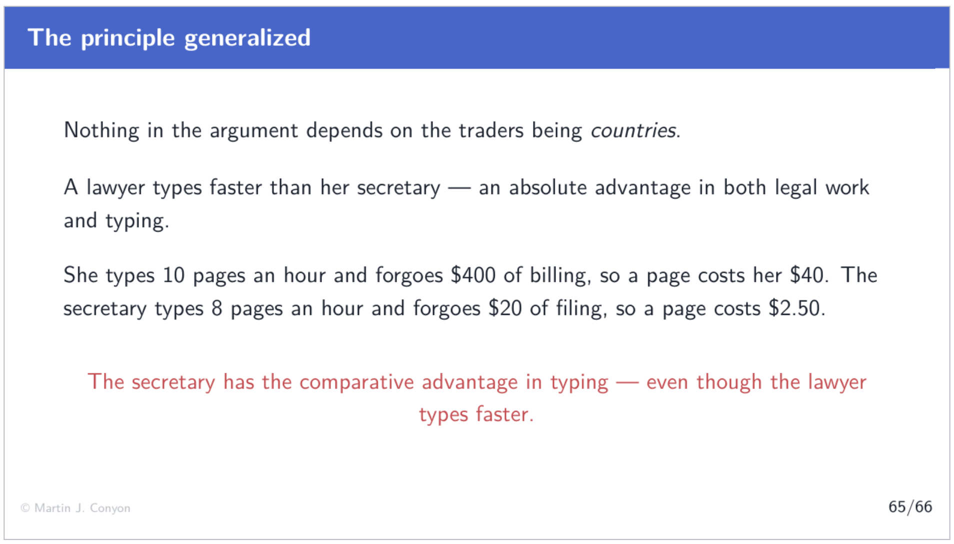
Put a number on it. The band was $0.89 < p < 1.20$, a width of about 0.31 cloth per wine. That is the whole advantage available to be shared. A shipping and tariff cost above roughly 0.31 cloth per wine leaves no price that beats autarky for both sides, and the trade does not happen.

So frictions do not merely shrink the gains. Past a threshold set by the difference in opportunity costs, they abolish them. Small comparative advantages are fragile; large ones survive high trade costs.

Comparative advantage is not fixed forever. It depends on *relative* productivity, and relative productivity is not a natural constant. It changes with capital investment, education, technology and the quality of institutions.

Which matters for how the result should be read. A country's comparative advantage describes where it stands now, not what it is condemned to. Countries that specialized in agriculture two generations ago now export electronics and software. The pattern of advantage is an outcome of choices as much as a constraint on them.

The principle generalized



Slide 65

Working through it

Nothing in the argument mentioned countries. It required only two producers, two activities, and different opportunity costs. Countries were Ricardo's example, not the theory's content.

The lawyer and the secretary. She types 10 pages an hour; he types 8. She has the absolute advantage in typing, and in legal work as well.

Work the opportunity costs, and note the unit. The comparison is per page, not per hour, because a page is what typing produces. An hour of her typing produces 10 pages and forgoes $400 of billing, so each page costs her $400/10 = \$40$. An hour of his typing produces

8 pages and forgoes \$20 of filing, so each page costs him $20/8 = \$2.50$.

So the secretary has the comparative advantage in typing — by a factor of sixteen — despite typing more slowly.

And the resolution is the same as Portugal's. Her hours are too valuable to spend on pages. Being faster at a task does not make doing it free; it only makes the hours it consumes more expensive.

The general statement. Wherever two parties give up different things to produce the same good, both do better when each does what it gives up least to do. That covers countries, firms, households, and how you spend an afternoon.